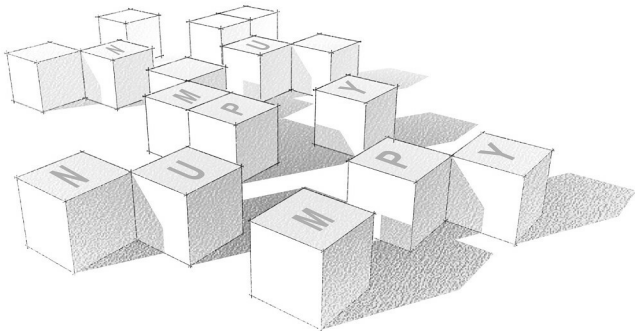


An Introduction to NumPy

Numerical Python or NumPy is a Python programming language library that supports large, multi-dimensional arrays and matrices, and comes with a vast collection of high-level mathematical functions to operate on these arrays.



Matrix-sig, a special interest group, was founded in 1995 with the aim of developing an array computing package compatible with the Python programming language. In the same year, Jim Hugunin developed a generalised matrix implementation package, Numeric. Later, in 2005, Travis Oliphant incorporated features of NumArray and C-API into Numeric code to create NumPy (Numerical Python), which was developed as a part of the SciPy project. These were separated from each other to avoid installing the large SciPy package just to get an array object.

In the rest of this article, '>>>' represents the Python interpreter prompt; statements without this prompt show the output of the code.

Installation

NumPy is a BSD-new licensed library for the Python programming language. It comes with Python distributions like Anaconda, Enthought Canopy and Pyzo. It can also be installed using package managers like dnf or pip. In Ubuntu and Debian systems, use the following code for installing NumPy:

```
sudo apt-get install python-numpy
```

Once NumPy is installed in the system, we need to import it to use the functionalities provided by it:

```
import numpy as n
```

The above command will create an alias 'n' while importing the package. Once you import the package, it will be active till you exit from the interpreter. If the programs are saved in files, you must import NumPy in each file.

NumPy, SciPy, Pandas and Scikit-learn

In Python, many packages provide support for scientific applications. NumPy, like Matlab, is used for efficient array computation. It also provides vectorised mathematical functions like *sin()* and *cos()*. SciPy assists us in scientific computing by providing methods for integration, interpolation, signal processing, statistics and linear algebra. Pandas helps in data analysis, statistics and visualisation. We use Scikit-learn when we want to train a machine learning algorithm. It seems like NumPy is inferior to all these packages. But the beauty is that all these packages use NumPy for their working.

ndarray

The soul of NumPy is 'ndarray', an n-dimensional array. You can see from the code segment given below that whenever you apply the function 'type' on any NumPy array, it will return the type *numpy.ndarray* irrespective of the type of data stored in it.

```
>>>import numpy as n
>>>a=n.array([1,2,3])
```